THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A group of 34 people has average weight 111 kg. Another group has average weight 72 kg. Together the average weight is 78 kg. The number of people in the second group is: 1

(a) 188 **(b)** 187 **(c)** 186 **(d)** 189

2. In what ratio must a 25% solution and a 41% solution be mixed to obtain a 34% solution? 1

(a) 7 : 9 **(b)** 7 : 10 **(c)** 9 : 7 **(d)** 8 : 9

3. A bus goes from A to B at 20 km/h and returns along the same road at 80 km/h. The average speed for the whole journey (km/h) is: 1

(a) 31 **(b)** 32 **(c)** 33 **(d)** 34

4. The average monthly salary of 22 employees is ₹66 thousand. An employee earning ₹25 thousand leaves and a new employee joins, and the average rises by ₹2 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 71 **(b)** 69 **(c)** 67 **(d)** 74

5. The average of 48 numbers is 75. The average of all 100 numbers (including 52 more numbers) is 88. The average of the 52 additional numbers is: 1

(a) 102 **(b)** 99 **(c)** 100 **(d)** 101

6. A shop earns a profit of 13% on sales of ₹2000 and 24% on sales of ₹9000. The overall profit percentage on total sales is: 1

(a) 21% **(b)** 23% **(c)** 22% **(d)** 24%

7. 750 mL of a 34% sugar solution is mixed with 1250 mL of a 66% sugar solution. The concentration of the mixture is: 1

(a) 53% **(b)** 55% **(c)** 54% **(d)** 56%

8. The average marks of 72 students is 90 and of another 144 students is 45. The average of all 216 students is: 1

(a) 62 **(b)** 59 **(c)** 61 **(d)** 60

9. **Assertion (A):** Class A has 20 students with average 60 marks and Class B has 30 students with average 70 marks; the combined average is 66. 1

Reason (R): The combined average equals the total of all marks divided by the total number of students.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The combined average of two groups is always the simple mean of the two group averages. 1

Reason (R): The combined average is the total of all the values divided by the total number of values.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A chemist has a 16% acid solution and a 64% acid solution. In what ratio must they be mixed to get a 36% solution? 2

12. A cyclist rides from home to a market at 30 km/h and returns by the same route at 45 km/h. Find the average speed for the whole trip. 2

13. The average monthly salary of 29 employees is ₹48 thousand. An employee earning ₹26 thousand leaves and a new employee joins; the average rises by ₹2 thousand. Find the new employee's monthly salary. 2

14. A chemist has a 52% acid solution and a 88% acid solution. In what ratio must they be mixed to get a 67% solution? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a class, 40 students have an average of 49 marks and the remaining students have an average of 90 marks. The class average is 70. Find the number of remaining students. 3

16. A final score combines Maths (score 80, weight 4), Science (score 78, weight 3), English (score 54, weight 8), Hindi (score 85, weight 2). Find the weighted average score. 3

17. The mean of 14 numbers is 51. Two more numbers 68 and 50 are added. Find the new mean and say whether it rises or falls. 3

OR

In a class, 78 students have an average of 71 marks and the remaining students have an average of 80 marks. The class average is 74. Find the number of remaining students. 3

18. A final score combines Maths (score 58, weight 3), Science (score 52, weight 5), English (score 51, weight 2), Hindi (score 55, weight 4). Find the weighted average score. 3

OR

The mean of 13 numbers is 63. Two more numbers 81 and 60 are added. Find the new mean and say whether it rises or falls. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In an annual test, Section A of Class 9 has 40 students with an average of 72 marks and Section B has 60 students with an average of 82 marks. 5

(i) Find the total marks of Section A.

(ii) Find the total marks of Section B.

(iii) Find the combined average of the two sections.

(iv) A new Section C of 50 students with average 90 is added. Find the new combined average of the three sections.

20. A trader mixes rice costing ₹145 per kg and rice costing ₹150 per kg in the ratio 2:3. Find the cost price per kg of the mixture and the selling price per kg to earn a profit of 25%. 5

OR

A family spent ₹3600 in January and ₹6000 in February. Food took 46% of the January total and 16% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks